

Where Next? Identifying U.S. Metros at Risk of Housing Unaffordability

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Abstract

This project builds an early-warning model trained on the historical pre-collapse trajectories of Tampa, Austin, and Boise to identify currently affordable metros following similar housing-market trajectories. Housing, income, and population data are joined by CBSA code to rank metros by affordability-risk resemblance. A high score signals a need for monitoring, not certain displacement.

Introduction & Motivation

Problem & question: Which housing-market and socioeconomic indicators best predict whether a mid-sized U.S. metro is following a pre-collapse affordability trajectory, and can those indicators identify at-risk metros before displacement pressures intensify?

Impact & stakeholders: City housing departments, regional lenders, planners, and real-estate stakeholders need lead time to monitor affordability stress and evaluate local supply constraints. Working- and middle-class residents in Tampa, Austin, and Boise experienced rapid housing-cost escalation. An interpretable early-warning tool can help stakeholders recognize similar patterns sooner.

Background & Related Work

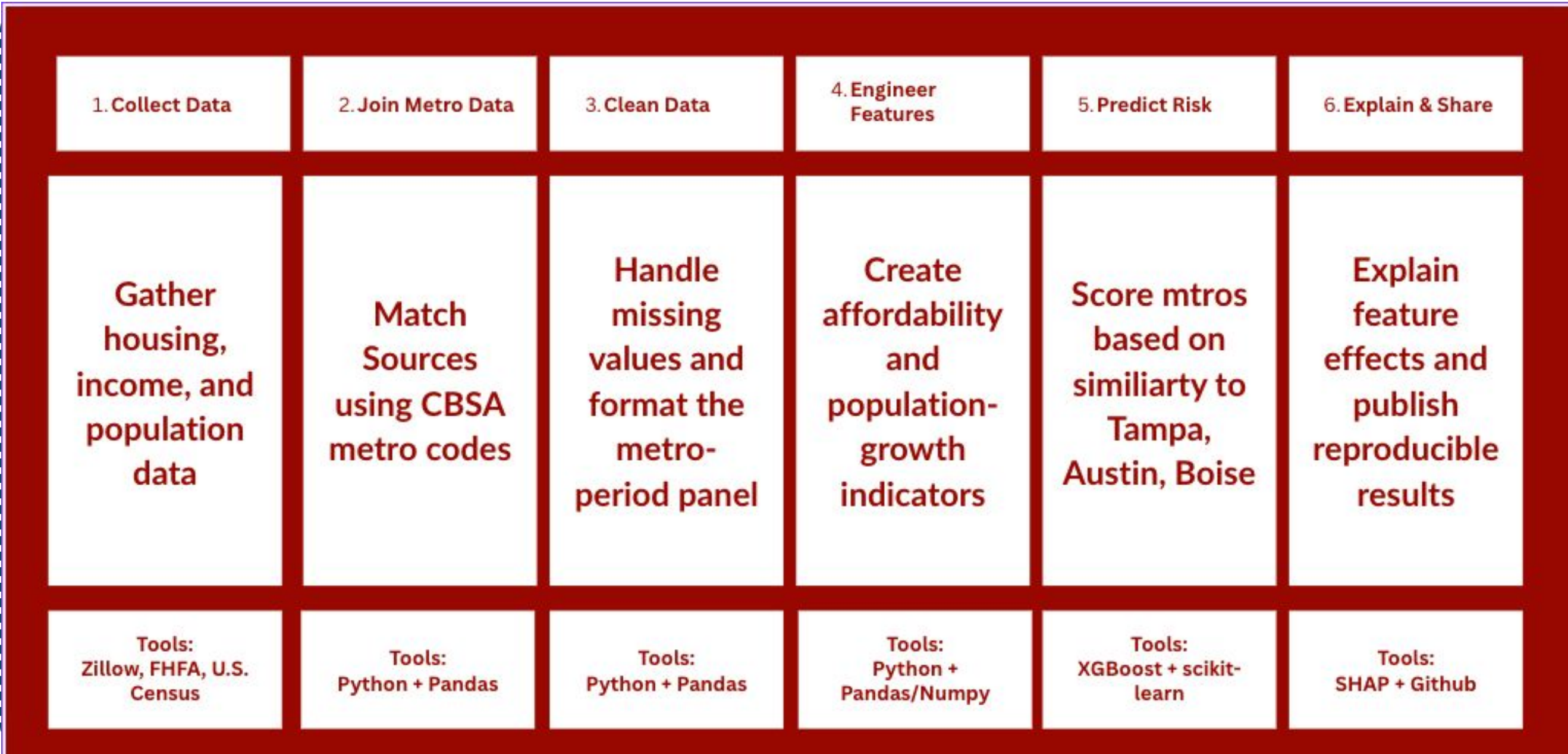
Zillow Research and the FHFA provide widely used measures of home values and price appreciation, but these resources primarily describe current and historical market conditions rather than flagging emerging affordability risk. Census data provide local income, housing-cost burden, demographic, and population context, but they are not typically integrated into a forward-looking metro-level risk score. This project combines these sources to treat Tampa, Austin, and Boise as historical affordability-collapse trajectories and test which pre-collapse signals also appear in other metros.

Data, Engineering & Ethics

Sources & scale: Zillow ZHVI, FHFA HPI, Census ACS, and Census Population Estimates are joined by CBSA code into a metro-period panel for feature engineering, model training, and risk scoring.

Pipeline: Python and R ingest, clean, standardize, and merge source data. The workflow includes quality checks, feature engineering, XGBoost scoring, SHAP interpretation, and reproducible GitHub storage.

Ethics & responsible use: Demographic variables are excluded from prediction and used only for post-hoc equity checks. Because false signals can influence investment or delay action, results are intended for monitoring—not deterministic forecasts or lending decisions.



Methods & Evaluation

Approach: An XGBoost classifier is trained on engineered affordability indicators: price-to-income ratio trend, affordability momentum, housing-supply pressure, mortgage burden relative to income, and population velocity. Tampa, Austin, and Boise serve as labeled affordability-collapse trajectories, and the trained model scores currently more affordable comparison metros by their similarity to those pre-collapse patterns.

Outcome definition: A metro is treated as entering an affordability-collapse trajectory when housing costs show sustained deterioration relative to local incomes according to the project's predefined price-to-income trend rule. The model identifies resemblance to this historical pattern; it does not estimate a causal effect or guarantee that a metro will become unaffordable.

Validation: Backtesting holds out the final two to three pre-collapse years for each training metro and tests whether the model would have identified elevated risk using only earlier observations. Performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC; cross-validation is used for model tuning and performance assessment across metro-period observations.

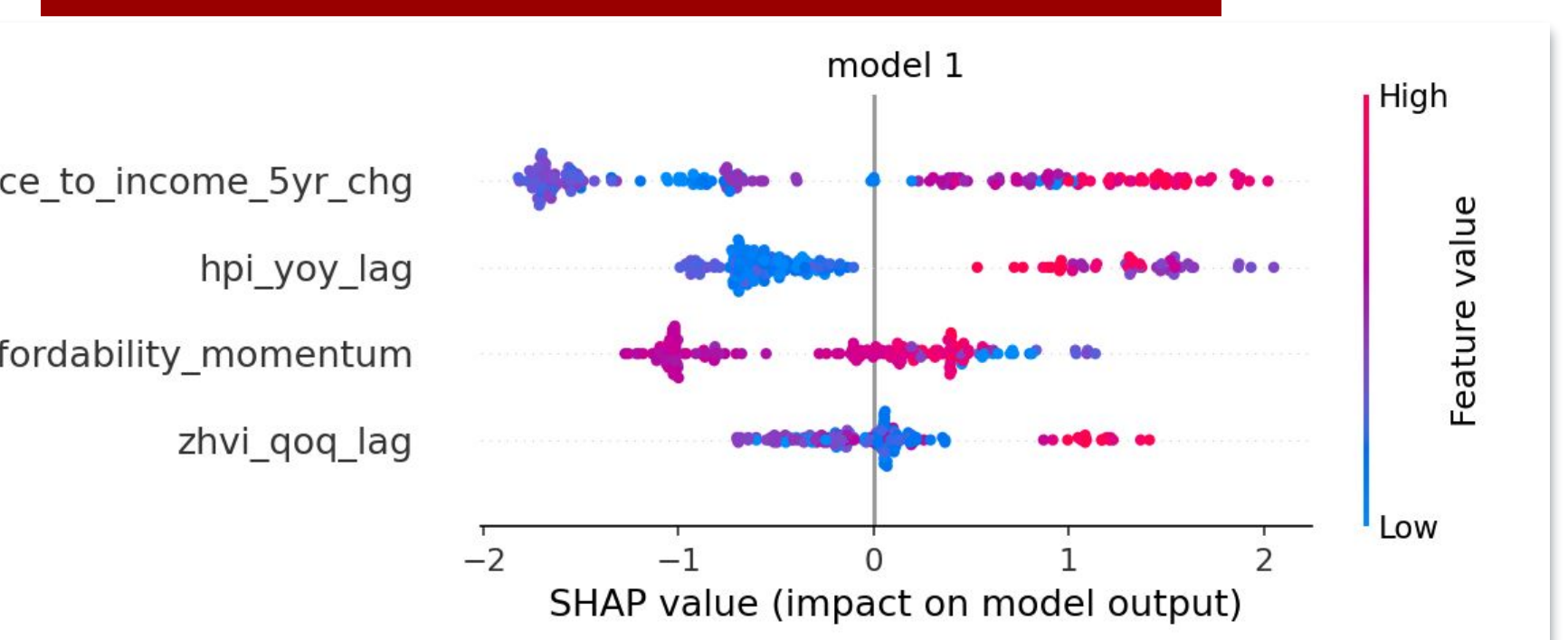
Results & Visualizations

Top 5 Holdout Metros by predicted Affordability Risk Score
Higher = stronger resemblance to Austin, Boise, pre-collapse pattern



Key finding: Bozeman, MT; Pinehurst-Southern Pines, NC; and Amherst Town-Northampton, MA most closely match the pre-collapse affordability trajectories of Tampa, Austin, and Boise. Higher risk scores indicate stronger similarity to those historical patterns and warrant closer monitoring rather than certainty of future affordability collapse.

SHAP Feature Effects on Affordability-Risk Predictions



Model performance: Five-year price-to-income change is the strongest driver of predicted affordability risk, with higher values generally pushing the model toward higher-risk classifications.

Discussion, Limitations & Conclusions

Discussion: A high risk score means a metro's current affordability trajectory resembles the patterns observed in Tampa, Austin, or Boise approximately three to five years before pronounced affordability deterioration. It does not mean that affordability collapse is inevitable or that the model identifies the cause of local housing-market change.

Limitations & future work: The three-city training set is small and geographically concentrated in high-growth Sun Belt and Mountain West markets, so results may generalize less reliably to older Rust Belt or Northeast metros. ACS measures can lag recent market conditions by roughly one year, meaning fast-moving changes may not yet appear in the data. Because the model is predictive and correlation-based, its rankings identify trajectory resemblance rather than causal mechanisms; false positives and false negatives remain possible.

Conclusion: Bozeman, Pinehurst-Southern Pines, and Amherst Town-Northampton show the strongest resemblance to the historical pre-collapse affordability trajectories of Tampa, Austin, and Boise. These metros warrant closer monitoring by housing authorities, planners, and lenders, including review of supply constraints, affordability trends, and potential displacement exposure. The model is intended to support earlier investigation and policy planning before affordability pressure fully materializes.

Reproducibility & References

Repository: [<https://github.com/BSmith-cpu/DATA-510-CAPSTONE->] · Key dependencies: Python 3.x, pandas, scikit-learn, XGBoost, SHAP
References: Zillow Research ZHVI methodology; FHFA House Price Index documentation, Urban Institute housing cost-burden reports

